Exp 27:Write a Prolog Program to implement Best First Search algorithm

Code:

```
edge(a, b).
edge(a, c).
edge(b, d).
edge(c, d).
edge(d, e).

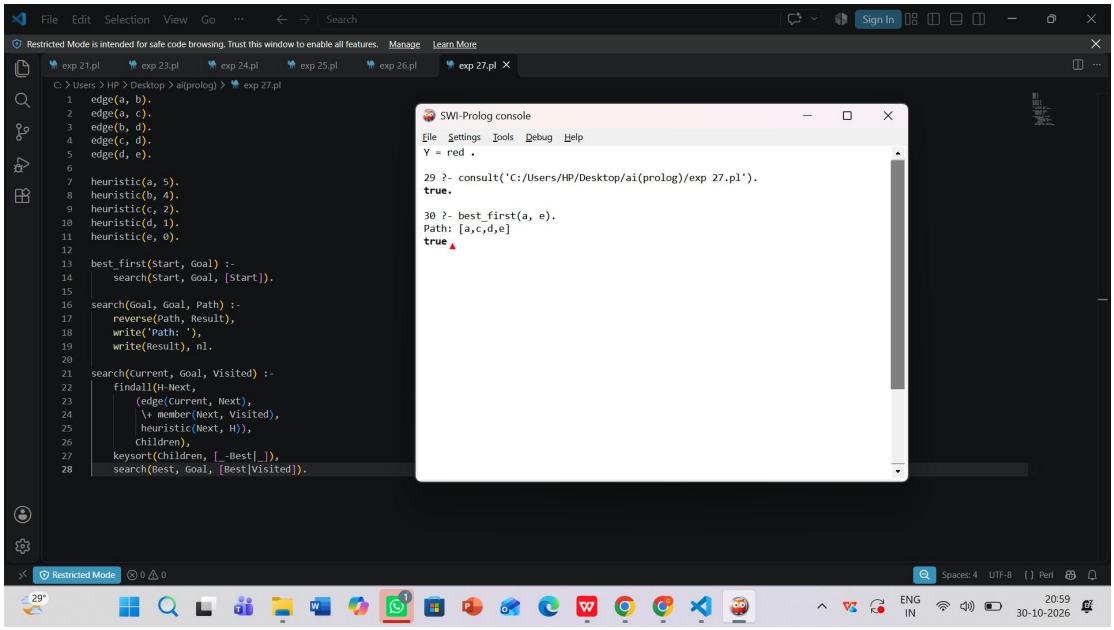
heuristic(a, 5).
heuristic(b, 4).
heuristic(c, 2).
heuristic(d, 1).
heuristic(e, 0).

best_first(Start, Goal) :-
    search(Start, Goal, [Start]).

search(Goal, Goal, Path) :-
    reverse(Path, Result),
    write('Path: '),
    write(Result), nl.

search(Current, Goal, Visited) :-
    findall(H-Next,
        (edge(Current, Next),
         \+ member(Next, Visited),
         heuristic(Next, H)),
        Children),
    keysort(Children, [_-Best|_]),
    search(Best, Goal, [Best|Visited]).
```

Output:


File Edit Selection View Go
Search
Sign In
Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More
exp 21.pl
exp 23.pl
exp 24.pl
exp 25.pl
exp 26.pl
exp 27.pl
C: > Users > HP > Desktop > ai(prolog) > exp 27.pl
edge(a, b).
edge(a, c).
edge(b, d).
edge(c, d).
edge(d, e).
heuristic(a, 5).
heuristic(b, 4).
heuristic(c, 2).
heuristic(d, 1).
heuristic(e, 0).
best_first(Start, Goal) :-
search(Start, Goal, [Start]).
search(Goal, Goal, Path) :-
reverse(Path, Result),
write('Path: '),
write(Result), nl.
search(Current, Goal, Visited) :-
findall(H-Next,
(edge(Current, Next),
\+ member(Next, Visited),
heuristic(Next, H)),
Children),
keysort(Children, [_-Best|_]),
search(Best, Goal, [Best|Visited]).
SWI-Prolog console
File Settings Tools Debug Help
Y = red .
29 ?- consult('C:/Users/HP/Desktop/ai(prolog)/exp 27.pl').
true.
30 ?- best_first(a, e).
Path: [a,c,d,e]
true
Restricted Mode
Spaces: 4 UTF-8 Perl
29°
ENG
IN
20:59
30-10-2026